

VWAP Reclaim Value Reclaim

Intraday-to-Swing Long Strategy for TradingView

Whitepaper

v2.2 Calibrated | TradingView Pine Script v6 | April 24, 2026

A professional documentation package for a long-only VWAP reclaim strategy. The model identifies intraday value recapture, adds calibrated regime filters, and manages trades with risk-based stops, scale-out logic, breakeven protection, and optional swing holds.

Prepared for strategy design, testing, and operational use. Educational and research material only; not financial advice.

Table of Contents

1. Executive Summary
2. Research Thesis
3. Strategy Architecture
4. Signal Specification
5. Filters and Calibration Logic
6. Risk Model and Trade Management
7. Backtesting Methodology
8. Operational Controls
9. Limitations and Governance
10. Appendix and References

1. Executive Summary

VWAP Reclaim Value Reclaim is a TradingView Pine v6 long-only strategy designed to detect a specific market behavior: price trades below the session value anchor, absorbs supply, and then reclaims that value area with enough evidence to justify a controlled long attempt. It is not a generic VWAP crossover system. It is a state-based setup engine that requires a prior dip, a reclaim event, risk validation, and optional confirmation from liquidity, volatility, trend, market proxy, and opening-range context.

The v2.2 Calibrated release solves the practical issue found in earlier strict versions: too many filters can produce no trades on short chart windows. The updated strategy therefore includes four operating profiles: Balanced - more trades, Signal finder/debug, Strict research, and Custom. Balanced is the recommended starting point for chart testing; Strict research restores the more conservative framework; Custom gives the operator full control over every gate.

Risk note: This model is a research and education tool. It does not predict future returns, guarantee profitability, or replace risk controls. Live deployment requires instrument-specific validation, broker reconciliation, and forward testing.

Design Goal	Practical Interpretation
Value reclaim	Enter only after the market loses session VWAP and then regains it, rather than buying an untested VWAP cross.
Calibrated confirmation	Use balanced default gates so the engine can produce testable signals before stricter rules are enabled.
Risk first	Reject entries whose proposed stop distance is too small, too large, or not tradable under the selected sizing mode.
Intraday-to-swing flexibility	Permit either end-of-day flattening or overnight continuation with a managed runner stop.
Diagnostics	Expose current blockers and historical raw reclaim blocks to explain zero-trade behavior.

2. Research Thesis

The strategy is built around a simple thesis: a reclaim of volume-weighted intraday value can become actionable when it occurs after a meaningful dip and is confirmed by trade quality. VWAP is widely used as a benchmark in institutional execution research and practice. Academic execution literature treats VWAP as a central benchmark for minimizing slippage and market impact, while retail and professional trading education commonly describe VWAP as an intraday reference for trend, liquidity, and value context.

However, the whitepaper does not assume that VWAP alone creates an edge. VWAP is a lagging, session-dependent measure. A raw cross above VWAP can be late, noisy, or overextended. The algorithm therefore separates three ideas: value anchor, reclaim evidence, and trade management. The entry engine is intentionally conservative about chasing price far above VWAP, and the exit engine is designed to protect capital if the reclaim fails.

Problem with naive VWAP crossover	v2.2 response
No proof of prior displacement	Requires a dip below VWAP within a lookback window.
Weak reclaim candles	Optional close-location and previous-high quality checks.
Chasing extended bars	Maximum close-above-VWAP threshold rejects stretched entries.
Flat or hostile regime	Optional trend, VWAP slope, ADX/DMI, RVOL, ATR%, gap, and proxy filters.
No risk discipline	Stop construction, risk percent gates, position sizing, TP1, breakeven, and trailing logic.
User confusion when no trades appear	Dashboard and BLOCK markers identify why raw reclaims were filtered.

Implementation note: The research improvement is not simply adding more indicators. It is adding explainable gates and profile controls so the trader can move from discovery to validation to stricter deployment.

3. Strategy Architecture

The v2.2 architecture is organized into six modules: session/value construction, opening-range context, reclaim event detection, regime filters, stop/sizing preparation, and trade management. Each module produces an explicit boolean gate. A long order can only be submitted when the selected operating profile permits the gate and the gate evaluates true.

Module	Primary Pine inputs	Purpose
Session/value	VWAP/value session, entry window, session days, source	Defines when VWAP accumulates and when entries are allowed.
Opening range	Opening range window, opening range filter	Adds early-session structure: off, above OR midpoint, or above OR high.
Reclaim engine	Entry mode, lookback, dip %, buffer %, max chase %, candle quality	Finds the dip-below and reclaim-back-above VWAP setup.
Regime filters	HTF EMA, proxy, VWAP slope, ADX/DMI, RVOL, ATR%, gap	Blocks trades in low-quality or hostile contexts when enabled.
Risk/sizing	Stop basis, ATR buffer, min/max initial risk %, sizing mode	Rejects untradeable risk and calculates optional risk-based quantity.
Exits	TP1, breakeven, runner trail, trend break, VWAP failure, carry overnight	Manages the position after fill.

High-level logic:

1. Build session VWAP from cumulative price-volume and volume.
2. Detect prior dip: $low \leq VWAP \times (1 - minDipPct)$.
3. Detect reclaim: close crosses back above VWAP inside the entry window.
4. Validate quality: buffer, no-chase, candle strength, optional previous-high break.
5. Validate context: optional opening range, trend, proxy, VWAP slope, ADX/DMI, RVOL, ATR%, gap.
6. Validate risk: stop distance must fit min/max risk percent and quantity rules.
7. Submit long order and attach stop/target exits.
8. Advance stop to breakeven and trail the runner after favorable movement.

4. Signal Specification

4.1 Session VWAP

The script calculates session VWAP manually rather than relying on an opaque chart setting. With the default source hlc3, price P is the average of high, low, and close for a bar. VWAP is:

$$VWAP_t = \frac{\sum(P_i \times Volume_i)}{\sum(Volume_i)}, \text{ for bars } i \text{ in the active VWAP session}$$

The strategy also calculates a volume-weighted variance and standard deviation to plot optional VWAP bands:

$$Variance_t = \frac{\sum(P_i^2 \times Volume_i)}{\sum(Volume_i)} - VWAP_t^2$$

$$UpperBand = VWAP + bandMult \times \sigma$$

$$LowerBand = VWAP - bandMult \times \sigma$$

4.2 Base Reclaim Event

The base event requires the market to show a prior loss of VWAP and then a close back above it. The raw setup is purposefully stateful: it remembers the most recent dip bar and optionally the most recent lower-band touch.

Condition	Meaning
dipEvent	The low touches or pierces VWAP by at least the configured minimum dip percent.
dippedEnough	The dip occurred within the reclaim lookback window.
crossedBackAbove	The current close is above VWAP and the prior close was not above VWAP.
belowDurationOk	The market spent at least the required number of bars below VWAP before reclaiming.
reclaimBufferOk	Close is above VWAP by the minimum reclaim buffer.
notChasing	Close is not too far above VWAP on the trigger bar.
candleQualityOk	Close is above open and high enough in the bar range.
breakPrevHighOk	Optional close above previous high to confirm short-term momentum.

4.3 Entry Modes

Immediate close reclaim enters when the base reclaim event and all active gates are true at the bar close. This is more responsive and is the default behavior for the Balanced and Signal finder profiles. **Retest-confirmed** arms the setup on the reclaim, waits for a controlled retest near VWAP that holds, and then requires price to break back above the reclaim trigger high. This mode is stricter and usually produces fewer trades.

5. Filters and Calibration Logic

Filters are not meant to be stacked blindly. v2.2 introduces profile-level overrides so a user can choose whether the model acts as a signal finder, a practical balanced strategy, a strict research strategy, or a fully custom engine.

Profile	Best use	Behavior
Balanced - more trades	Initial chart testing and practical research	Keeps RVOL and ATR regime but relaxes trend, OR, VWAP slope, and quality gates that caused zero-trade behavior in v2.1.
Signal finder/debug	Proving that VWAP reclaim events exist on the chart	Loosens most entry restrictions and allows more entries per session. Not intended as a final live setting.
Strict research	Conservative hypothesis testing	Re-enables the original research gates such as HTF trend, VWAP slope, ADX/DMI, RVOL, and ATR regime.
Custom	Instrument-specific model design	Uses the individual input switches exactly as configured by the operator.

5.1 Regime Filter Rationale

Filter	Rationale	Common failure mode
HTF EMA trend	Avoid long entries when the higher timeframe is below its EMA.	Can block valid intraday reversals; disabled by default in Balanced.
Market proxy	Require broad market support for single-name longs, such as QQQ or SPY context.	May be irrelevant for non-US equities, futures, crypto, or idiosyncratic movers.
VWAP slope	Require session value to flatten or rise, implying improving intraday acceptance.	Can be too slow after sharp V-shaped recoveries.
ADX/DMI	Seek sufficient trend strength and optional positive directional movement.	Can reject early reversals before trend strength expands.
RVOL	Require participation; avoids thin, low-conviction crosses.	May reject quiet but orderly reclaim setups.
ATR%	Avoid dead volatility and extremely high-volatility traps.	Needs symbol-specific calibration.
Gap filter	Avoid chasing extreme session gaps or weak gap-down contexts.	Can block news-driven continuation days.

Implementation note: The dashboard displays the current blocker. BLOCK markers on the chart show historical raw reclaim events that were detected but filtered out.

6. Risk Model and Trade Management

The model treats risk as a first-class input. A reclaim is not tradable unless the proposed stop produces a positive risk distance and falls within the min/max initial risk percent bands. This avoids entries with stops that are too tight to survive normal noise or too wide to offer acceptable reward-to-risk.

Component	Logic
Initial stop basis	Choices: reclaim bar low, swing low, lower VWAP band, or structure + ATR.
ATR buffer	Subtracts a multiple of ATR from the selected stop anchor to reduce stop clustering.
Min/max risk %	Rejects trades whose stop distance is outside the configured risk envelope.
Risk-based sizing	Optional quantity = capital at risk / dollar risk per share or contract.
TP1 scale-out	Optionally exits part of the trade at a fixed R multiple.

Component	Logic
Breakeven trigger	Moves stop to entry once price reaches the configured R threshold.
Runner trail	ATR chandelier, VWAP/ATR hybrid, or structure swing.
Signal exits	Optional end-of-day flat, max hold bars, HTF EMA loss, or two closes below VWAP.

The default runner trail is VWAP/ATR hybrid. After breakeven is armed, it compares an ATR-based chandelier stop to a VWAP-minus-ATR-buffer stop and keeps the more protective value. Since the stop is advanced with `math.max()`, it is designed as a non-loosening trailing stop for long positions.

7. Backtesting Methodology

Strategy Tester results are simulations. TradingView strategies can place hypothetical orders and show simulated performance; the broker emulator, strategy properties, commission, slippage, limit fill assumptions, and Bar Magnifier settings can materially change results. The whitepaper recommends a staged protocol rather than immediate optimization.

Stage	Objective	Pass/fail question
Smoke test	Confirm that the script loads on an intraday chart and produces signals in Signal finder/debug.	Does the engine detect raw VWAP reclaims?
Balanced baseline	Test Balanced - more trades over at least several months.	Does the setup produce enough trades for analysis?
Filter attribution	Turn on one filter at a time in Custom.	Which filters improve risk-adjusted results and which only reduce sample size?
Walk-forward	Tune on one period and validate on a different period.	Does performance persist out of sample?
Forward test	Run paper or tiny-size live monitoring.	Do fills, slippage, and behavior match the backtest assumptions?

Recommended metrics include total trades, average R, profit factor, max drawdown, exposure time, average adverse excursion, average favorable excursion, win rate by time of day, results by volatility regime, and filter-block attribution.

Risk note: Avoid declaring success from a one-week test window. VWAP reclaim setups are regime-sensitive and require broad sampling across trend days, range days, gap days, earnings/news periods, and low-volume sessions.

8. Operational Controls

The strategy includes practical controls for the issues most likely to cause user confusion: invalid chart timeframe, too many active filters, small sample windows, and chart display settings. The soft guard allows the script to load on daily charts but prevents entries unless the chart is intraday and higher-timeframe relationships are valid.

Control	What it does
Soft timeframe guard	Prevents entries unless the chart is intraday; displays a dashboard warning instead of throwing a runtime error.
Main blocker row	Summarizes the current reason a trade cannot trigger.
BLOCK marker	Marks raw reclaim bars that were blocked by one or more filters or risk gates.
Max entries per session	Limits churn and overtrading during noisy VWAP oscillations.
Carry overnight toggle	Switches between day-trade behavior and intraday-to-swing behavior.
Alerts	Supports long signal alerts and retest-armed alerts.

9. Limitations and Governance

The strategy should be governed like a research model, not like a prediction oracle. Its logic is transparent, but transparent rules can still be overfit. Because VWAP is based on historical intraday price and volume, it is lagging. Because the model is long-only, it may underperform during bearish or low-liquidity regimes. Because Pine strategies use simulated orders, results

may diverge from broker fills, especially in fast markets.

- Use liquid symbols and realistic slippage/commission assumptions.
- Do not optimize every input at once. Isolate one hypothesis at a time.
- Keep a written version log of settings changes and test periods.
- Separate discovery settings from deployment settings.
- Forward test before any live capital deployment.
- Treat news, earnings, halts, and abnormal spreads as separate risk conditions.

10. Appendix: Core Gate Map

Gate name	v2.2 meaning
chartOk	Intraday chart and valid higher/proxy timeframe relationships.
sessionOk	Inside VWAP session and entry window, with session entry count below max.
baseReclaimSignal	Raw reclaim plus active quality, context, and regime gates.
retestConfirmSignal	Retest mode has armed, held VWAP, and broken trigger high.
riskOk	Initial stop distance is positive and within configured risk limits.
qtyOk	Risk-based quantity is valid when risk-based sizing is selected.
longSignal	Final order condition: chartOk, setup, flat position, riskOk, and qtyOk.

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